NoteLMs: A Three-Model Organization Framework for Student Notes

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Abstract

Every student knows about the chaotic, disorganized notes they take in class each day out of habit, only to never review them again. Existing note organizers include Microsoft OneNote, Apple Notes, and, worst of all, physical paper notes. All of these require users to organize and format their notes manually, adding to the chore of note-taking.

NoteLMs lets users paste in plain-text notes or upload images of handwritten notes. It automatically classifies the subject of the notes, places them in a corresponding subject folder, and fully formats them in HTML so they are easy to read in the future. This lets students focus on class instead of organization. For review, Voice uses a natural-sounding AI model to read notes aloud. Behind the scenes, we use three language models to classify notes and evaluate the models' accuracy in real time, contributing data to the research portion of this project.

1 Introduction

NoteLMs (notelms.com) is a website that automatically sorts and formats students' notes while also contributing research data to the underlying project.

- Are LMs capable of categorizing notes by subject?
- If so, which LM performs best?

After thousands of evaluation runs, we found that language models are capable of categorizing notes by subject with an accuracy of over 90%. The best LM we tested was a version of BERT fine-tuned on 16,000 pieces of note-subject data. In this paper, we introduce NoteLMs as a product and then examine the underlying evaluation data.

2 Related Work

The three main note-taking solutions students use are physical notebooks, Microsoft OneNote, and Apple Notes. Malone's seminal study of how people organize their physical desks

found that most individuals rely on random, unstructured piles rather than organized filing. This shows that, long before digital tools existed, users struggled to maintain organized, categorized notes[1].

With the introduction of digital note-taking tools such as Microsoft OneNote and Apple Notes, this sense of disorganization has continued. These programs expect users to set up and organize note file systems by themselves.[2][3]

All current note-taking solutions require manual organization and formatting.

3 Conceptual Solution

Our solution automatically organizes and formats the notes that a user uploads. There are two possible inputs: raw text notes typed by the user or an image of handwritten notes. There are several outputs, depending on which stage of learning the user is in:

- A subject classification of the notes, which causes each note to be sorted into the corresponding folder.
- Notes formatted in HTML and/or JSON.
- A natural-sounding voice reading pasted text aloud.

Each note also receives an automatic summary for quick context. Finally, NoteLMs has an LLM chatbot built into the site that can answer questions about the notes within the user's account.

4 Implementation

NoteLMs is a fully functional website live at notelms.com. The web application UI is hosted on Vercel, while the backend connects to my local M3 Ultra Mac Studio through a Cloudflare Tunnel.

Most AI processing is done locally on the Mac Studio, especially the operation of the three main models used by the site: zero-shot BERT, fine-tuned BERT, and gpt-oss-20b. The exception occurs when a user uploads an image. In that case, we send the image through the OpenAI API to first extract its text before feeding that text through the rest of the local pipeline. Both BERT models are run on PyTorch, and gpt-oss-20b is run on LMStudio.

To train and fine-tune the BERT model for subject classification, we gathered 16,000 pieces of notes-subject pairs from online, subject-specific textbooks. We trained BERT on all 16,000 pieces of data, equally distributed across subjects. Because BERT is such a small model, we were able to perform full-parameter fine-tuning.

[1] T. Malone, "How Do People Organize Their Desks? Implications for the Design of Office Information Systems," *ACM Transactions on Office Information Systems*, vol. 1, no. 1, pp. 99–112, 1983.

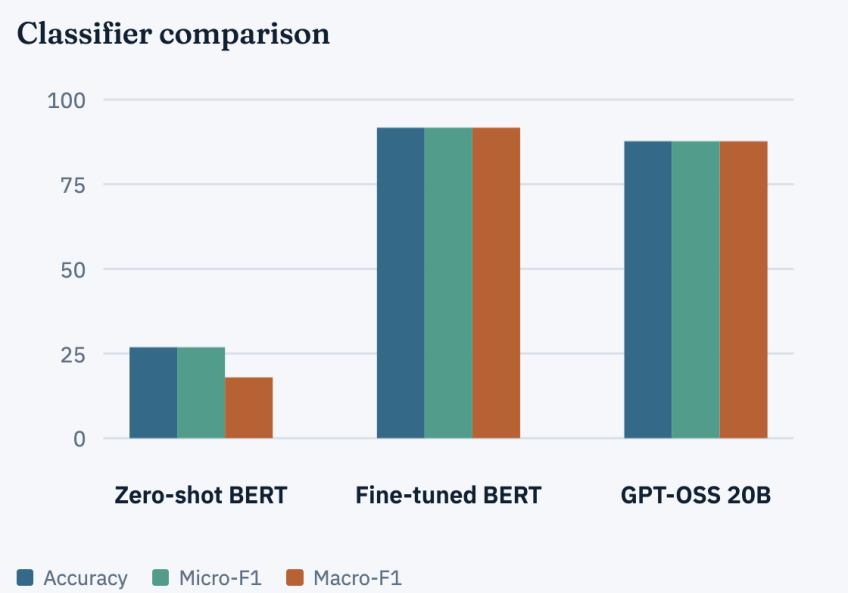
[2] "How I Decluttered 4,000 Apple Notes in 15 Minutes a Week," *MacOClock, Medium*, 2026.

[3] "OneNote sections not staying in order," *Microsoft Q&A*, 2020.

5 Evaluation

Once we finished training and fine-tuning BERT, we tested it against the other two models in their default configurations. Those models were zero-shot BERT and default gpt-oss-20b. Our test suite consisted of another 2,000 textbook note-subject pairs, and we ensured that none of them appeared in the original training data.

As shown in Figure 1, we found that the best model was fine-tuned BERT, with an accuracy of 91.7%. Our leading theories for gpt-oss-20b's slightly lower accuracy are that it overthinks problems during reasoning, and that it's just a general model rather than a fine-tuned one.



SYSTEM	ACCURACY	MICRO-F1	MACRO-F1
Zero-shot BERT	26.9%	26.9%	17.9%
Fine-tuned BERT	91.7%	91.7%	91.7%
GPT-OSS 20B	87.7%	87.7%	87.7%

Figure 1: Accuracy of the 3 LMs tested for note-subject classification: zero-shot BERT, fine-tuned BERT, and gpt-oss-20b.

A challenge is translating evaluation accuracy into real-world accuracy. We solved this with the orchestrator model, which reviews the three LMs' classifications and decides which one to use. It uses each model's typical accuracy to select a final classification.

Another possible issue with our evaluations is the lack of diverse data. This is addressed by using user data: every time a user uploads notes to NoteLMs, we use that data to run our three models and score them based on the user's final subject selection.

6 Conclusion

At the end of the day, the problem we set out to solve was giving students access to a tool that automatically organizes their notes. Based on our >90% success rate for individual models in evaluations, combined with the orchestrator model, we are confident that NoteLMs delivers this experience. Future work would involve implementing local image-to-text processing and testing a wider variety of reasoning LLMs, rather than just gpt-oss-20b.

References

[1] Malone, T. W. "How Do People Organize Their Desks? Implications for the Design of Office Information Systems." *ACM Transactions on Office Information Systems* 1, no. 1 (1983): 99–112.

[2] MacOClock. "How I Decluttered 4,000 Apple Notes in 15 Minutes a Week." *Medium*, 2026.

[3] Microsoft Q&A. "OneNote Sections Not Staying in Order." 2020.